Exponential growth and decay

When we talked earlier about using the logistic equation to model population growth, we mentioned briefly that we sometimes use the exponential equation

$$P(t) = P_0 e^{kt}$$

to model a population that grows exponentially. This equation is usually only used to model populations that are very small, since the exponential nature of the growth means the population needs to be fairly unrestricted by its environment.

What we want to say now is that this equation is actually the solution to the initial value problem

$$\frac{dP}{dt} = kP, \qquad P(t_0) = P_0$$

When we use this exponential growth and decay equation to model population, we'll often use the variables P and t, where the derivative dP/dt models the rate at which the population P changes over time t.

The general growth and decay equation

But we can also write this differential equation more generally as

$$\frac{dx}{dt} = kx, \qquad x(t_0) = x_0$$

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and in this general form, we'll use this initial value problem all the time to model real-world phenomena like bacterial growth, compound interest, and half life of a decaying substance.

We call k the **constant of proportionality**. If $k > 0$ we call it a growth constant, whereas if $k < 0$ we call it a decay constant. So for applications like population growth and compound interest, k will be positive, but for decay applications like half life, k will be negative.

This growth/decay differential equation is so useful because it tells us that the rate of growth is proportional to the population/amount/etc. In other words, the larger the population, the faster is replicates, or the more money, the faster it grows. And the equation is a separable equation,

$$\frac{dx}{dt} = kx$$

$$\frac{1}{x}\, dx = k\, dt$$

which means that we already know how to find the solution. Remember that separable equations are linear equations, so we could solve this as a first order linear equation by finding the integrating factor, or we can solve it this way as a separable equation:

$$\int \frac{1}{x}\, dx = \int k\, dt$$

$$\ln|x| = kt + C$$

$$e^{\ln|x|} = e^{kt+C}$$

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$$|x| = e^{kt}e^{C}$$

$$x = Ce^{kt}$$

Since the general solution will always be the same, whenever we get this kind of initial value problem with $dx/dt = kx$ and $x(t_0) = x_0$, we can go straight to the solution equation to plug in the initial condition. Substituting the initial condition is what allows us to solve for the value of C.

Let's do an example where we use this exponential growth equation to model the growth of a bacteria population.

Example

The size of a bacteria population is initially P_0, and the population has grown to $3P_0$ after 2 hours. If the growth rate is proportional to the size of the population at any time t, how long did it take for the population to double?

The problem gives us the initial conditions $P(0) = P_0$ and $P(2) = 3P_0$. Substituting the first condition into the general solution equation lets us solve for the constant C.

$$P = Ce^{kt}$$

$$P_0 = Ce^{k(0)}$$

$$P_0 = C$$

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Now we'll substitute $C = P_0$ and $P(2) = 3P_0$ into the general solution equation to find k, the constant of proportionality.

$$P = P_0 e^{kt}$$

$$3P_0 = P_0 e^{k(2)}$$

$$3 = e^{2k}$$

Apply the natural log to both sides.

$$\ln 3 = \ln(e^{2k})$$

$$\ln 3 = 2k$$

$$k = \frac{\ln 3}{2}$$

Now we can use k to find the time it took for the population to double, or the time it took to reach a population of $2P_0$.

$$2P_0 = P_0 e^{\frac{\ln 3}{2}t}$$

$$2 = e^{\frac{\ln 3}{2}t}$$

$$\ln 2 = \ln e^{\frac{\ln 3}{2}t}$$

$$\ln 2 = \frac{\ln 3}{2}t$$

$$t = \frac{2 \ln 2}{\ln 3}$$

$$t \approx 1.26$$

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This result tells us that the bacteria doubled the size of its original population in about 1.26 hours, or about 1 hour and 16 minutes.

If we check our result from the last example against the original question, it should make sense to us. It took 2 hours for the population to triple, which means it should take somewhere between 1 and 2 hours for the population to double, since the growth rate is always increasing. And in fact, our result of 1.26 hours for the population to double matches that intuition.

Let's look at one more example, this time with decay instead of growth, where the constant of proportionality is negative.

Example

A certain isotope decays to $25\,\%$ of its original amount after 50 years. If the rate of decay is proportional to the amount remaining at any time t, how much of the isotope will remain after 75 years?

The problem gives us the initial conditions $x(0) = x_0$ and $x(50) = (1/4)x_0$, so we'll substitute the first into the general solution equation to find C.

$$x = Ce^{kt}$$

$$x_0 = Ce^{k(0)}$$

$$x_0 = C$$

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Now substituting $C = x_0$ and $x(50) = (1/4)x_0$ will let us find k.

$$\frac{1}{4}x_0 = x_0 e^{k(50)}$$

$$\frac{1}{4} = e^{50k}$$

$$\ln \frac{1}{4} = \ln(e^{50k})$$

$$\ln 1 - \ln 4 = 50k$$

$$-\ln 4 = 50k$$

$$k = -\frac{\ln 4}{50}$$

Finally, we'll substitute this decay constant and $t = 75$ to find the amount of the isotope that remains after 75 years.

$$x(t) = x_0 e^{-\frac{\ln 4}{50}(75)}$$

$$x(t) = x_0 e^{\ln(4^{-\frac{3}{2}})}$$

$$x(t) = \frac{1}{4^{\frac{3}{2}}}x_0$$

$$x(t) = \frac{1}{8}x_0$$

So $1/8$th of the original amount of the isotope remains after 75 years of decay.

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